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AI Engineering

Agentic Dependency Parsing in Low-Resource Settings

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Abstract

In natural language processing (NLP), dependency parsing refers to the task of constructing sentence-level dependency trees. Often, the resulting dependency–head relations are annotated to specify the type of dependency. Large collections of such dependency trees—so called treebanks—are a valuable resource in the study of syntax, grammar, and typology. The manual creation of such dependency trees, however, is a very laborious undertaking which should, if possible, be performed at least semi-automatically. For high-resource languages, many popular NLP libraries (Stanza, NLTK, SpaCy) provide the necessary tooling for exactly this sort of (pre-)annotation. In the case of low-resource languages, however, such tools are typically not available. This observation has instigated us to test whether an AI agent, i.e. an LLM equipped with adequate tools, can function as a general purpose dependency parser, thereby eliminating the need to train purpose-built neural networks as used by the aforementioned libraries.

1 Introduction

In natural language processing (NLP), dependency parsing refers to the task of constructing sentence-level dependency trees. Syntactically, a dependency tree is a directed, connected, acyclic graphs with words as nodes and *dependency-head*-relations as edges. Semantically, it reflects a sentence's dependency structure: Each word is linked to the head of the phrase it is part of. There is always exactly one word that does not itself have a head. This word is called the *root* of the sentence. The edges of such a dependency tree are often annotated with a label taken from a pre-defined set of possible dependency relations.

2 Related Work

Dependency parsing: Stanza, SpaCy, NLTK.

Cross-lingual transfer: (García-Ferrero)

3 Our Approach

4 Experiment

5 Results

6 Conclusion

7 Author Contribution Statement

Everyone explains the extend of their contribution in their own words.

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Bibliography

García-Ferrero, Iker. "Cross-Lingual Transfer for Low-Resource Natural Language Processing." 2025, https://arxiv.org/abs/2502.02722.